



Regime-Gated PCA Microstructure Flow

A Cross-Sectional Residual-Flow Strategy on Crypto Perpetual Futures

QF635 Market Microstructure and Algorithmic Trading

Project Report - Guideline-Aligned Final Draft

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1. Executive Summary

This project researches, implements, and validates a market-neutral cross-sectional strategy on Binance USD-M crypto perpetual futures. The objective is to test whether idiosyncratic taker-flow pressure, after removing common crypto-wide flow with rolling PCA, can predict short-horizon relative returns across altcoin perpetuals. Bitcoin is used mainly as a market-regime reference, while the traded universe consists of nine liquid altcoin perpetuals.

The strategy combines three components. First, a daily BTC regime gate classifies the market into Strong, Moderate, or Risk-Off states. Second, a one-minute PCA residual-flow alpha isolates asset-specific order-flow pressure and trades it cross-sectionally. Third, a microstructure confirmation layer uses order-book imbalance, microprice deviation, spread filters, and liquidity screening before target weights are constructed. The portfolio is designed to be dollar-neutral, volatility-scaled, capped at the single-name level, and rebalanced at short intervals.

The main empirical result is that the signal exists, but the current liquidity-taking implementation is not investable. The residual-flow signal has a one-minute Spearman information coefficient of about -0.029 with a t-statistic near -16, and the IC remains negative from 1 to 30 minutes. This is evidence of short-horizon mean reversion. When traded in reversal direction with full deployment and zero trading cost, the strategy earns about +6.49% over thirty days with a daily Sharpe near 5.7. However, the alpha horizon is so short that turnover overwhelms the edge once realistic costs are applied. The estimated break-even cost is only about 0.05 bp, far below plausible fee, spread, and slippage levels for a taker strategy.

The second important finding is that the BTC regime gate is poorly matched to a market-neutral cross-sectional book. In the evaluation window, it leaves the strategy out of the market for roughly 90% of bars and does not improve the economics of the residual-flow signal. The all-gates version loses 0.50% over the month, underperforming cash, although it avoids most of the drawdown in the equal-weight long-only altcoin basket, which falls 18.76%. That relative outperformance should be interpreted as downside avoidance, not extracted alpha.

Final assessment: do not deploy the current strategy with real capital as a liquidity-taking strategy. The research signal is meaningful, but the execution model is wrong. The natural next step is to reuse the residual-flow signal inside a market-making or quote-skew framework that earns spread rather than paying it, or to search for a slower-horizon signal with similar IC and materially lower turnover.

2. Motivation and Market Opportunity

Crypto perpetual futures are attractive for market microstructure research because they trade continuously, publish rich intraday data, and attract a wide mix of retail, proprietary, arbitrage, and liquidity-providing participants. The market exposes taker-buy volume, taker-sell volume, funding rates, spreads, and order-book information at high frequency. These features make short-horizon order-flow effects plausible.

The opportunity is based on temporary dislocations rather than long-term crypto direction. When aggressive buying or selling is concentrated in a single altcoin, that pressure can push the asset away from the rest of the cross-section. If the move is mostly liquidity demand rather than durable information, the next few minutes should show partial reversal. This is the economic intuition behind using idiosyncratic taker-flow residuals as a mean-reversion signal.

The central challenge is that raw crypto order flow is contaminated by market-wide movement. Many altcoins move together when Bitcoin drives broad risk appetite. A raw flow signal can therefore mistake market beta for alpha. PCA residualization is used to remove common crypto-wide flow and focus on the residual component that is more likely to represent asset-specific imbalance. Funding crowdedness adds a slower positioning prior, while order-book confirmation attempts to filter weak or illiquid entries.

The opportunity may persist because crypto markets still contain fast-moving liquidity demand, retail overreaction, segmented participants, and variation in liquidity across instruments. However, persistence is not enough. The opportunity must also be tradeable after transaction costs, which becomes the key commercial test of the project.

3. Market Microstructure Analysis

The traded instruments are perpetual futures. They have no expiry, use funding payments to anchor futures prices to spot markets, and trade continuously. Price discovery is highly linked to Bitcoin and broader crypto risk, but individual altcoins also experience asset-specific flow shocks. The strategy is built around separating those two components.

The main flow primitive is one-minute Taker-Flow Imbalance, defined for each asset i and minute t as:

$$TFI_{i,t} = (TakerBuyVolume_{i,t} - TakerSellVolume_{i,t}) / (TakerBuyVolume_{i,t} + TakerSellVolume_{i,t})$$

A positive value indicates net aggressive buying; a negative value indicates net aggressive selling. On its own, TFI is not enough because common market flow can dominate. Rolling PCA is therefore applied to the cross-section of TFI series over a 120-bar window. The first principal component approximates broad crypto-wide flow, and the second component captures secondary cross-sectional rotation.

Each asset's TFI is then residualized against the first two principal components using a 60-bar rolling beta window:

$$TFI_{i,t} = \beta_{i,1} PC1_t + \beta_{i,2} PC2_t + \epsilon_{i,t}$$

The residual $\epsilon_{i,t}$ is the idiosyncratic flow pressure used for alpha. On the test sample, the first principal component explains only about 14% of variation, which suggests that residual idiosyncratic flow remains economically meaningful. In live and paper trading, this signal is supplemented with order-book imbalance, microprice deviation, spread filters, and an ADV20 liquidity screen. A limitation is that historical Level-2 order-book data are not available through the Binance REST interface used here, so book confirmation is only approximated in the historical backtest.

4. Strategy Research and Signal Development

The trading hypothesis is that PCA-neutralized residual flow mean-reverts at short horizons. In other words, an asset with unusually strong idiosyncratic aggressive buying should underperform the cross-section over the next few minutes, while an asset with unusually strong idiosyncratic aggressive selling should outperform. The project did not assume this direction in advance; the direction was tested through out-of-sample information coefficients.

Signal / feature	What it measures	Why it is included	Trading role
Residual TFI	Asset-specific aggressive flow after removing common PCA factors	Separates idiosyncratic imbalance from broad crypto beta	Primary alpha signal
Funding signal	Crowded long/short pressure embedded in perp funding	Adds slower positioning information	Secondary alpha input
Order-book imbalance	Depth imbalance near best bid/ask	Checks whether book pressure confirms the flow signal	Entry filter
Microprice deviation	Best-quote pressure relative to mid price	Captures near-touch short-term pressure	Entry filter
Spread and ADV20	Trading cost and liquidity conditions	Avoids weak or expensive fills	Eligibility filter

Table 1. Feature design and contribution to trading decisions.

The alpha is constructed by cross-sectionally standardizing the residual flow and blending it with the funding signal:

$$\text{Alpha}_{i,t} = 0.8 * Z(\text{ResidualFlow}_{i,t}) + 0.2 * \text{FundingSignal}_{i,t}$$

Signals are ranked cross-sectionally. Q80/Q20 selection is used rather than fixed thresholds because quantile selection is less sensitive to scale drift across assets and naturally creates a balanced long-short structure. The evidence supports reversal, not continuation: ICs are negative at 1, 2, 5, 10, and 30 minutes, with the strongest effect at the one-minute horizon.

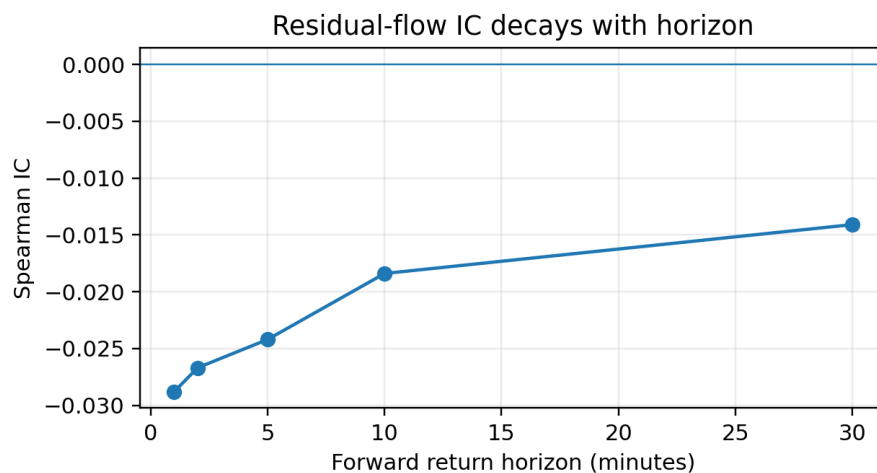


Figure 1. Residual-flow IC is negative and decays smoothly with horizon, indicating short-horizon reversal.

5. Model Selection and Justification

The selected model is deliberately simple and interpretable. It uses rolling PCA for common-factor removal, rolling OLS for residualization, cross-sectional z-scores for normalization, and rank-based quantile selection for portfolio formation. The prediction target is short-horizon relative return across the altcoin cross-section.

Several alternatives were considered. Raw TFI was rejected because it is too exposed to common market movement. Fixed thresholds were rejected because signal scale changes over time and across assets. Continuation trading was rejected because realized IC shows the opposite sign. A stricter IC stability gate was tested to reduce reliance on unstable sign estimates. Avellaneda-Stoikov market-making was also considered; it is likely the more natural execution framework for this signal, but it changes the project from a taker-style long-short strategy into a passive quoting model.

The final model was chosen because it is robust, explainable, and aligned with the cross-sectional market-neutral objective. Its weakness is not lack of statistical signal. Its weakness is that the alpha horizon is too fast relative to liquidity-taking costs. This is exactly why a more complex predictive model would not solve the main problem unless it also reduces turnover or improves execution economics.

6. Strategy Implementation Logic

The implemented strategy operates in three layers. First, a BTC daily regime gate classifies market state using five-day and twenty-day momentum together with twenty-day realized volatility against a 252-day Q80 volatility threshold. Strong regimes receive 1.0x gross exposure, Moderate regimes receive 0.5x, and Risk-Off receives 0x.

Second, the one-minute residual-flow alpha ranks eligible altcoin perpetuals. In reversal mode, assets with the highest positive residual-flow pressure are candidates for short positions, while assets with the most negative residual-flow pressure are candidates for long positions. Third, microstructure confirmation checks order-book imbalance, microprice deviation, spread, and liquidity conditions before weights are formed.

Entry occurs when an asset survives liquidity screening, ranks into the long or short selection quantile, and passes the relevant confirmation filters. Exit occurs when the asset drops out of the selected quantile, fails the filter, reaches a rebalance event with a lower target weight, or when the regime/risk layer moves the strategy to zero exposure. Position sizing is volatility-scaled and capped at the single-name level. The implementation requires both long and short legs to survive confirmation; otherwise it sets the portfolio to zero rather than allowing an unintended one-sided book.

Execution is modeled as target-weight rebalancing in the backtest. Turnover is charged through the cost parameter, and funding is booked at settlement timestamps. This is a useful research approximation but not a full fill simulator.

7. Trading System Architecture

The system was implemented as a complete event-driven trading framework rather than a single research notebook. The main modules are data retrieval, feature construction, signal generation, decision-making, risk checks, order management, execution, position tracking, and monitoring/reporting.



Figure 2. End-to-end trading system architecture from market data to feedback.

The shared alpha engine spans `pca_flow.py`, `microstructure_confirmation.py`, and `combined.py`. It is used consistently by the historical backtester, live framework, and user interface. Live execution through `broker.LiveBroker` and

binance_rest.py includes signed REST orders, tick and step rounding, GTX post-only handling, asynchronous user-data fill processing, cancel/replace logic, listenKey keepalive, and testnet operation by default.

The architecture evolved through five stages: exchange onboarding, runnable signal pipeline, risk and execution integration, discovery of consistency issues, and pre-submission cleanup. This process matters because several improvements were not alpha changes, but corrections to system semantics.

From Research Signal to Trading System

A 5-Stage Evolution Timeline

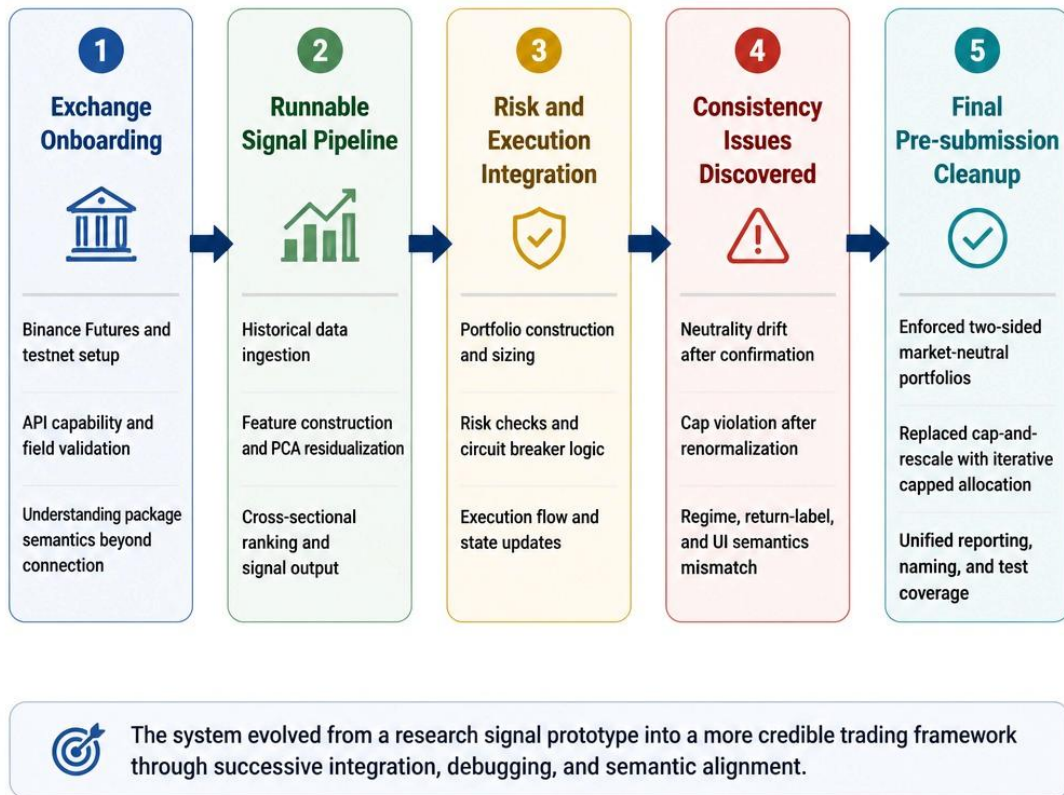


Figure 3. Development path from research signal to submission-ready trading framework.

8. Risk Management Framework

Risk management is embedded at both the portfolio and pre-trade layers. The portfolio is designed to maintain $\sum(w)=0$, apply volatility scaling, respect single-name caps, and scale gross exposure by regime. The final implementation replaces one-shot cap-and-rescale logic with an iterative capped allocation procedure, preventing post-normalization cap violations.

Risk area	Implemented control	Purpose / limitation
Position exposure	Single-name cap, gross exposure cap, market-neutral weights	Controls concentration and directional exposure
Volatility	Volatility-scaled weights and BTC regime scaling	Reduces exposure in higher-risk market states
Drawdown	Daily-loss circuit breaker at -3%	Halts strategy under acute losses
Order risk	Per-order quantity cap, reduce-only handling, rate limit	Prevents oversized or repeated orders
Operational risk	Fail-fast credential and venue checks, kill switch	Avoids silent unsafe live trading
VaR / margin ratio	Not fully implemented in this submission	Should be added before real production deployment

Table 2. Risk controls and remaining production gaps.

If the strategy behaves unexpectedly, losses are controlled by exposure limits, drawdown halts, circuit breakers, liquidity filters, and neutralization rules. However, a production version should add real-time VaR, exchange margin-ratio monitoring, reconciliation alerts, latency monitoring, and explicit soft/hard risk thresholds beyond the current research framework.

9. Backtesting Methodology

The backtest is an event-by-event replay over real Binance one-minute klines, daily Bitcoin candles, and funding history for 19 May to 18 June 2026. The run can be reproduced with: `python backtest.py --start 2026-05-19 --days 30`.

Look-ahead bias is controlled by using only completed prior daily candles for the regime filter and fitting PCA only on past rolling windows. PnL is computed on a weights basis: target weights are assumed to be achieved at each bar, turnover is charged through a configurable cost parameter, and funding is applied at settlement timestamps. This setup is appropriate for alpha diagnosis, but it does not model partial fills, queue priority, maker/taker uncertainty, or order latency.

Transaction costs are central to the analysis. The strategy is evaluated at zero cost to measure gross signal value, and at positive cost levels to test whether the signal can survive realistic implementation frictions. Because historical Level-2 books are unavailable, the order-book confirmation layer is approximated historically and should be treated as unvalidated until tested in paper trading or live-style replay. The cleanest signal diagnostic is therefore the cross-sectional Spearman IC of raw residual-flow alpha against forward returns, which is independent of regime, portfolio overlays, and costs.

10. Performance Analysis

The evaluation window contains 43,200 one-minute bars, and the IC diagnostic covers approximately 43,020 periods. The residual-flow IC is statistically strong, negative, and monotonic across horizons. That pattern is consistent with short-horizon mean reversion rather than continuation.

Horizon	IC	t-stat	Hit %
1 min	-0.0288	-16.6	45.9
2 min	-0.0267	-15.5	46.1
5 min	-0.0242	-14.2	46.4
10 min	-0.0184	-10.8	47.0
30 min	-0.0141	-8.3	47.7

Table 3. Information coefficient of residual-flow signal versus forward returns.

The all-gates strategy loses slightly after costs and trades very little. Since the strategy is continuously weighted and market-neutral, the most meaningful activity statistic is not conventional trade count but rebalancing activity and turnover. The backtest records 122 rebalances, average turnover of 0.351 per rebalance, and total turnover of about 42.8x the book.

Metric	Value
Total return (30d)	-0.50%
Annualized return	-5.91%
Annualized volatility	0.97%
Sharpe (annualized, from daily P&L)	-5.03
Maximum drawdown	-0.53%
Win rate (in-market bars)	43.0%
Number of trading events	122 rebalances; total turnover approx. 42.8x
Total trading cost	0.43%
Time in market	1.0%
Performance by regime	Risk-Off 90%, Moderate 10%, Strong 0%
Equal-weight 9-alt benchmark	-18.76%
Excess vs long-only benchmark	+18.26%

Table 4. Full-strategy performance with all gates enabled.

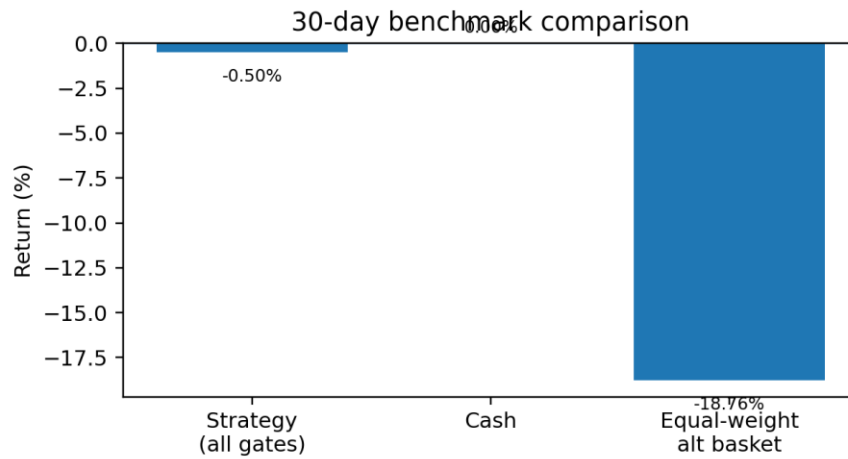


Figure 4. The strategy protects capital versus the falling altcoin basket, but underperforms cash.

The correct benchmark interpretation is conservative. Against cash, the strategy underperforms by 0.50%, so the all-gates version is not investable. Against the equal-weight altcoin basket, it outperforms by 18.26%, but this mostly reflects neutrality and inactivity during an altcoin drawdown. It should be described as downside avoidance rather than alpha capture.

11. Strategy Optimisation

Optimisation focused on whether the strategy could preserve the residual-flow edge while reducing turnover and cost. The most important changes and tests were: direction selection by realized IC, removal or weakening of the BTC regime gate, rebalance-frequency changes, signal smoothing, no-trade bands, microstructure confirmation, and capped portfolio allocation.

The largest performance driver is direction. Continuation trading has the wrong sign; reversal trading is correct. The second-largest driver is the regime gate. Removing it increases deployment from roughly 1% to almost the full sample and reveals a strong gross signal. However, that improvement does not solve the cost problem. Slower rebalancing and heavier smoothing lower turnover but also destroy gross alpha because the edge is concentrated at short horizons.

Configuration	Change tested	Impact
Direction: continuation to reversal	Flip sign according to negative IC	Turns signal from wrong-way to profitable gross direction
Remove BTC regime gate	Allow neutral book to trade in more bars	Raises gross alpha but increases turnover exposure
Longer rebalance interval	Rebalance every 30-60 minutes	Cuts cost but weakens alpha materially
Smoothing	Average signals across longer windows	Reduces noise but dilutes fast reversal edge
Strict neutrality after confirmation	Require both long and short legs	Prevents unintended one-sided portfolios
Iterative capped allocation	Enforce cap after normalization	Fixes cap violation after rescaling

Table 5. Optimisation changes and their effect on strategy quality.

12. Robustness Testing and Validation

Robustness testing gives a mixed but useful conclusion. The IC evidence is statistically strong in the main May-June window and has the same sign across multiple horizons, which reduces the chance that the result is only one-horizon noise. However, the signal is not stable enough to justify hardcoding the direction forever: a March 2026 window shows a weaker one-minute IC near -0.010, close to the noise floor.

Configuration	Time in market	Gross	Cost @1bp	Net @1bp
Full strategy: regime + IC gate on	1.0%	approx. +0.0%	0.4%	-0.50%
No regime, reversal, no confirm, cost 0	99.6%	+6.49%	0	+6.49%
No regime, reversal, no confirm, cost 1bp	99.6%	+6.49%	approx. 126%	approx. -120%
+ rebalance 30, smooth 15	99.6%	+0.50%	19.3%	-18.84%
+ rebalance 60, smooth 1	99.6%	+2.63%	10.7%	-8.11%
+ rebalance 60, smooth 30	99.6%	-1.29%	9.0%	-10.31%

Table 6. Robustness and ablation tests under different gates, cost assumptions, and rebalance settings.

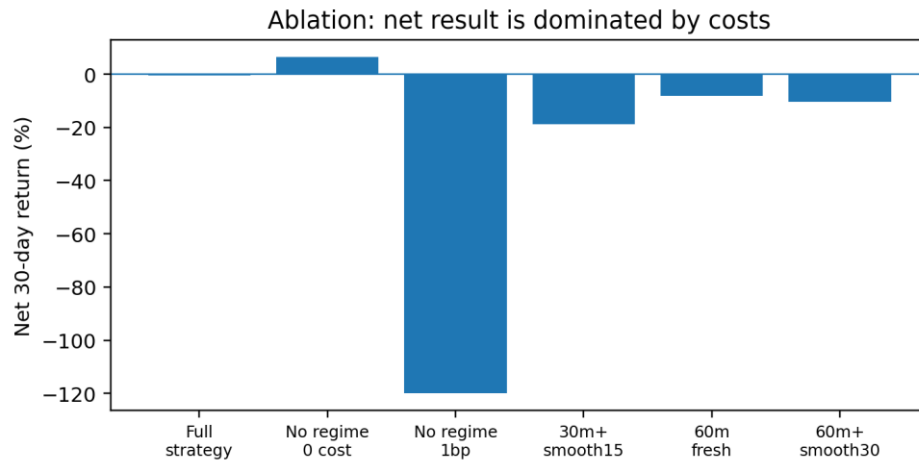


Figure 5. Cost sensitivity dominates the gross edge in liquidity-taking configurations.

The main robustness conclusion is that the signal is real but not robustly monetizable under a taker execution model. The break-even cost is about 0.05 bp for the fast configuration and only improves to about 0.24 bp at hourly rebalancing, still far below realistic costs. This is not a parameter-tuning issue. It is a structural mismatch between alpha horizon and execution model.

13. Production Deployment Considerations

The current system is suitable for continued research, paper trading, and testnet validation. It is not production-proven. A production deployment would require hardened market-data infrastructure, stronger exchange reconciliation, real-time OMS state checks, trade/position break alerts, margin and liquidation monitoring, latency monitoring, automated kill switches, and post-trade reporting.

The production recommendation is not to deploy the current liquidity-taking strategy. The more defensible path is to convert residual-flow alpha into a market-making or quote-skew signal. In that framework, the model would not cross the spread to chase a one-minute edge. Instead, it would adjust passive quotes, quote size, or skew according to expected short-horizon reversal, attempting to earn spread while reducing adverse selection.

Before real deployment, the next validation sequence should be: testnet dry-run, live paper trading with real-time order-book capture, maker/taker fill analysis, queue-position estimation, reconciliation testing, capital-limited shadow mode, and only then a small controlled production pilot if the cost-adjusted results remain credible.

14. Limitations and Future Enhancements

The project has important limitations, and they materially affect deployment judgment.

1. The strategy is uneconomical as a liquidity taker. The estimated break-even cost of about 0.05 bp is far below realistic fees, spreads, and slippage.
2. The sample window is short. The main results cover one thirty-day period, and the IC weakens in at least one earlier subperiod.
3. Historical Level-2 book data are unavailable in the backtest. The microstructure confirmation layer is only approximated and must be validated through paper/live data capture.
4. The backtest is weights-based. It does not simulate queue priority, partial fills, latency, adverse selection, or maker/taker uncertainty.
5. The BTC regime gate is not well matched to a market-neutral cross-sectional book and should not be treated as validated.
6. Risk infrastructure is incomplete for real production. Real-time VaR, margin-ratio monitoring, liquidation buffers, and reconciliation alerts should be added.

Future enhancements should focus on execution model rather than cosmetic parameter tuning. The most promising direction is a market-making or quote-skew version using captured Level-2 data. A second direction is to search for slower-horizon residual-flow features with lower turnover. A third direction is broader cross-window validation across different crypto regimes and liquidity environments.

15. Conclusion

The strategy should not be deployed with real capital in its current specification. The research produced a meaningful result: PCA-neutralized idiosyncratic taker-flow contains short-horizon mean-reversion information. The signal is statistically strong in the main window, economically intuitive, and aligned with market microstructure logic. However, it is too fast to trade profitably by taking liquidity.

The project therefore succeeds as a professional trading-system study, not as a live trading recommendation. It shows the full path from market opportunity and signal design to implementation, risk control, backtesting, validation, and deployment judgment. It also shows why realistic costs and system details matter: a strong gross signal can still be commercially unusable if turnover and execution frictions dominate.

The final assessment is conservative. The current strategy is a solid research and paper-trading framework. The next version should either become a passive quote-skew/market-making model or identify a slower signal with similar predictive quality and lower turnover. Until that happens, real-capital deployment is not justified.